PVEMON

PVE node and backup monitor

User guide

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What it is and what you can do

PVEMon is a read-only monitor for your Proxmox VE servers: nodes, virtual machines, LXC containers, Docker, disks and backups. It shows the state at a glance and warns you when something fails. It never changes anything on your servers.

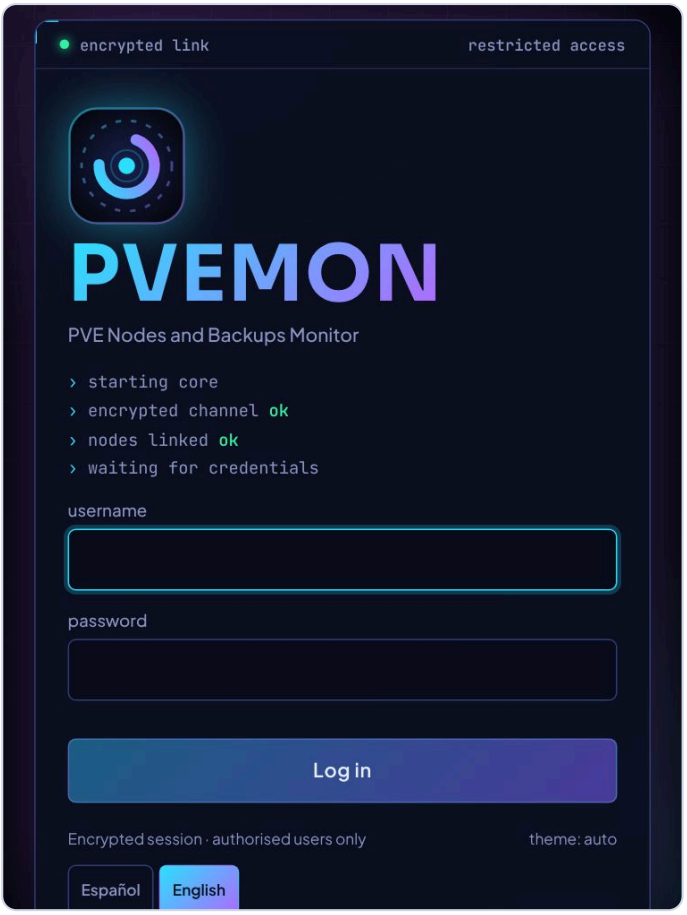
- **Dashboard** with CPU, memory and disk rings for every node and machine, and warnings about what needs attention.
- **Backups**: Proxmox (vzdump) backups, NAS and other programs (by e-mail), file checks on a network drive and PVEMon's own backup.
- **Alerts** by e-mail, Telegram, push, webhook and Microsoft Teams; weekly summary.
- **Disk health** (SMART, wear, temperature, ZFS).
- **Kiosks** for fixed screens, and **installation as an app** on phone or computer.

> If you only need the backup edition, there is **BCKMon**, with its own guide.

PVEMon and BCKMon share the same code and database: every improvement reaches both.

Signing in to PVEMon

Open the address your administrator gave you, type your username and password and press **Sign in**. If your account has two-step verification (2FA) it will also ask for the code from your authenticator app. Below the form you can choose the language (Español / English).



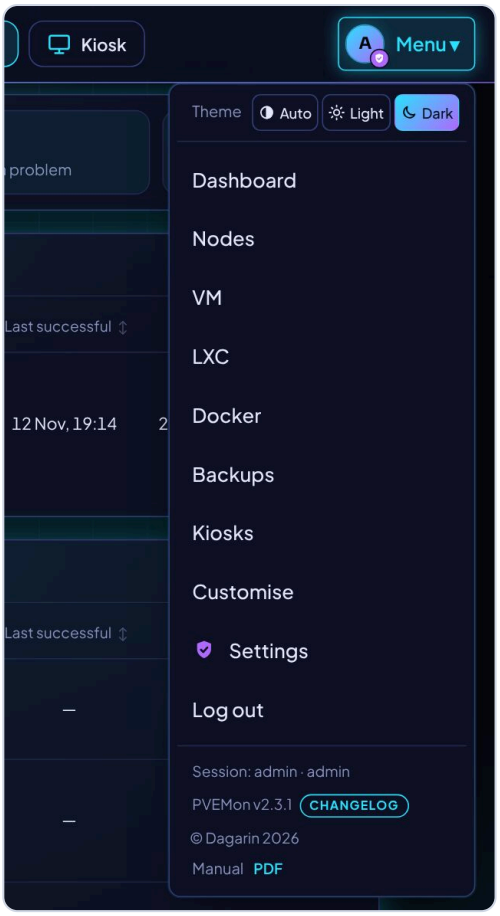
Sign-in screen

> After 10 failed attempts in 15 minutes the account is locked for a while. If you forget your password, ask an administrator to reset it.

Title bar and menu

The header has three areas: the name and icon on the left; **in the centre**, the navigation buttons; and on the right the alert badge (if any) and the **Menu** button.

- **Status**: the main dashboard.
- **Backups**: the backups panel.
- **Kiosk**: opens a kiosk full screen.



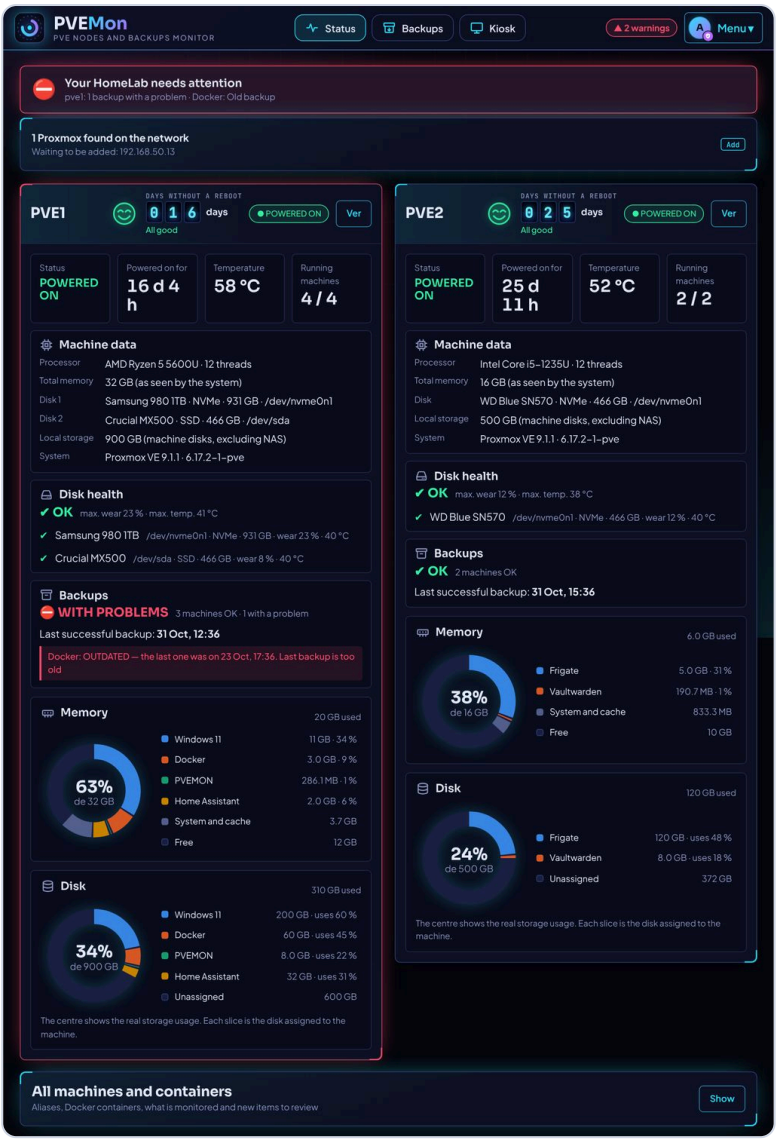
The menu: theme, options, Log out, version and manual

The **Menu** contains, from top to bottom:

- **Theme** (Auto, Light, Dark), on a single line, at the very top.
- The screens: **Dashboard, Nodes, VM, LXC, Docker, Backups, Kiosks, Customise** and **Settings** (administrators). A monitor user only sees Alerts and Backups besides the common ones.
- **Log out**, right below the options.
- Your session, the version and the **ChangeLog** button (history of changes).
- **Manual · PDF**: downloads this guide in your language.

The Dashboard

The **Dashboard** is the main screen: on top, the warnings that need attention; below, one card per node with CPU, memory and disk rings, state, uptime and temperature; then the machines and containers of each node. Click any of them to open its detail.



The Dashboard

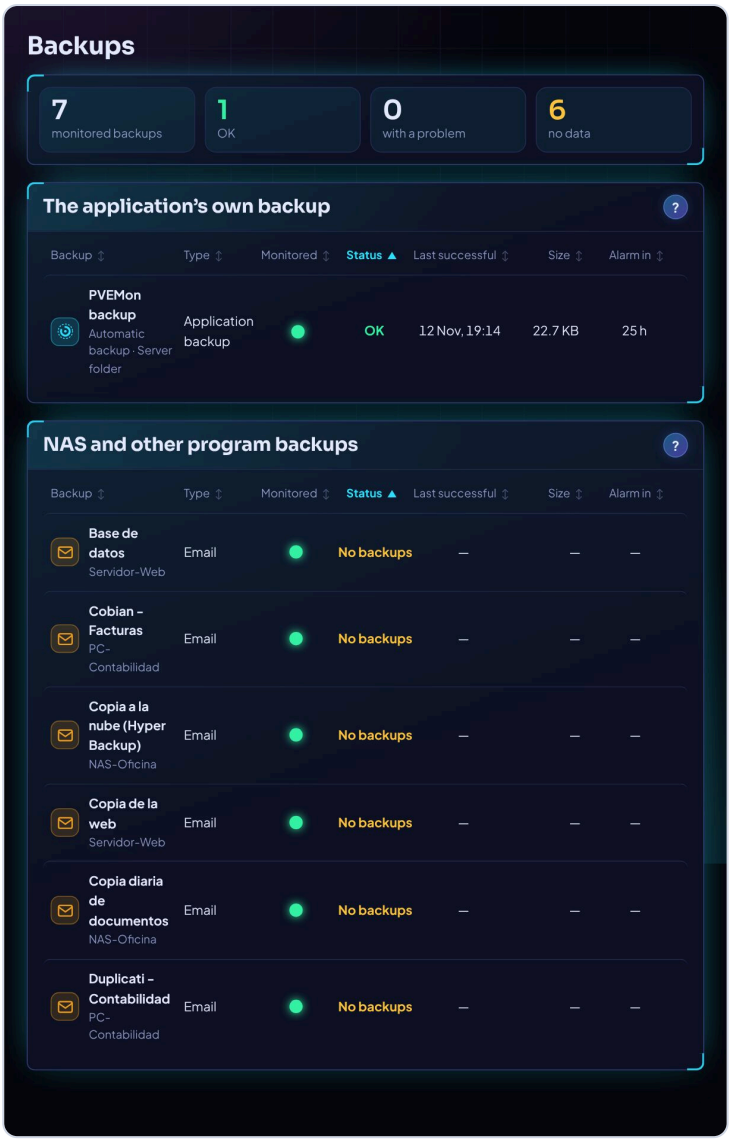


Machine detail

Backups screen

This is the panel for all backups. At the top, the summary: **monitored, correct, with a problem** and, if any, **no data**. Below, one card per type of backup:

- **The application's own backup**: PVEMon's automatic backup (always first), with its state, date of the last successful backup, size and alarm.
- **Proxmox backups**: the vzdump backups of your machines and containers.
- **NAS and other programs**: those arriving by e-mail (Synology Hyper Backup, Duplicati, Cobian...) and the file checks on a network drive.



Backups screen

Column	What it shows
Backup	Name, source (machine or NAS) and an icon per type.
Type	E-mail, Proxmox, file on a network drive or application backup.
Monitored	Green dot: it is watched and warns. Red: it is not watched.
State	Correct, Error, Old or No data; with the reason if there is a problem.
Last successful	Date and time of the last good backup.
Size	How much that backup took.
Alarm in	Time left before the warning if no other good backup arrives («overdue» if already past).

You can **sort** by any column by clicking its heading (on a phone, with the «Sort by» drop-down). Click a row to open its sheet: the state on top and, next to it, a **face with a counter** that changes face and colour as the alarm date approaches (and goes into crisis if the backup failed or the date has

passed); below, last successful backup, size, last attempt and limit. The application-backup row takes you to its settings.



Backup sheet, with the face and the counter towards the alarm

> If a backup is red even though the last one was successful, read the reason: it usually means more time than allowed has passed without another backup.

Alerts

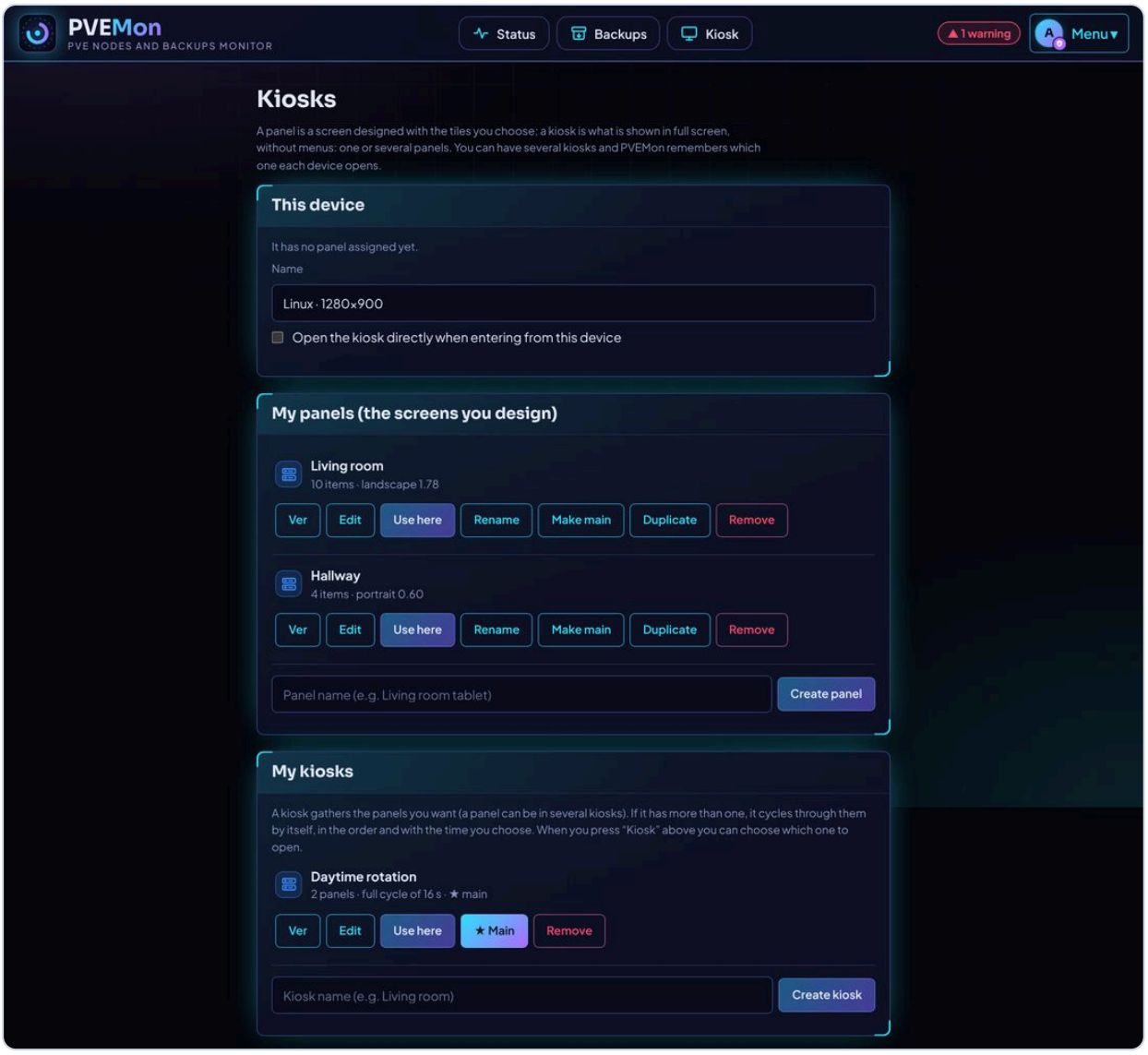
When something needs attention the red button in the header counts the warnings and the **Alerts** screen lists them with their history. An alert fires when the condition lasts the configured time and resolves itself after a minute of stability.

You receive notices through the channels the administrator configured (e-mail, Telegram, push, webhook, Teams). In **Customise** you choose which ones you want for yourself.

Kiosk

A kiosk is a read-only screen meant for a TV or fixed monitor: it shows the panels you choose, with no menus, and can rotate between groups of panels. It opens with the **Kiosk** button in the header (full screen) and you leave it with the Esc key.

In **Kiosks** (menu) you create and edit your own with a canvas editor: you add panels (summaries, backup state, text), position and size them, and can undo and redo. You can have several kiosks and mark one as the main one.



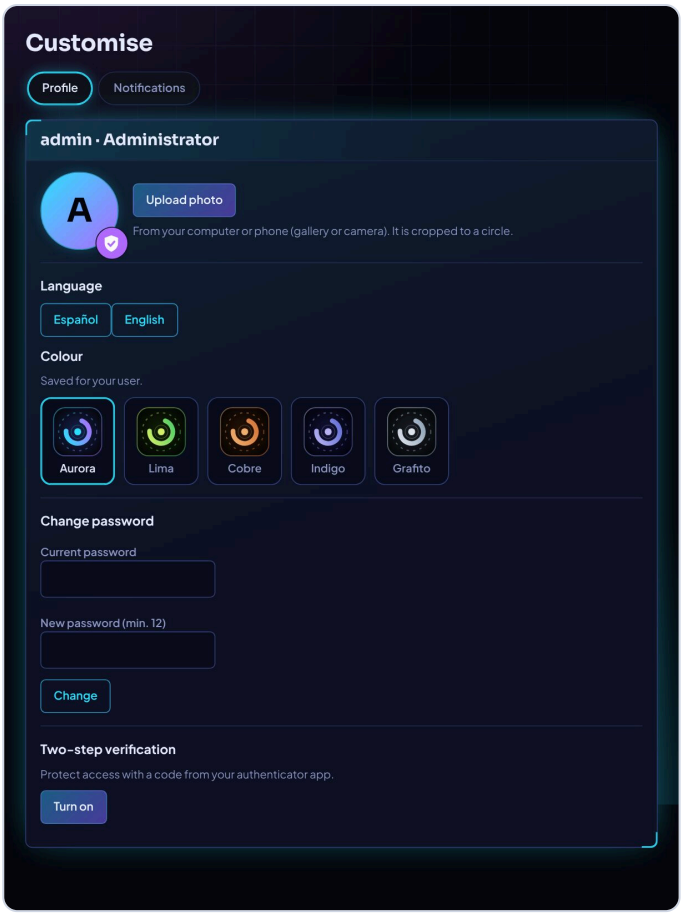
Kiosks screen



A kiosk on screen

Customising your account

Customise gathers your own settings: your photo (cropped to a circle), your name, the language, the password change (asks for the current and the new one, minimum 12 characters), two-step verification (2FA, with an authenticator app and a QR code) and **My notices**, where you choose which channels to use.



Customise

Installing PVEMon as an app

- **iPhone/iPad (Safari)**: Share → Add to Home Screen.
- **Android (Chrome)**: ⋮ menu → Install app.
- **Computer (Edge/Chrome)**: install icon in the address bar.

Once installed it receives push notifications with its own icon.

Privacy and access limits

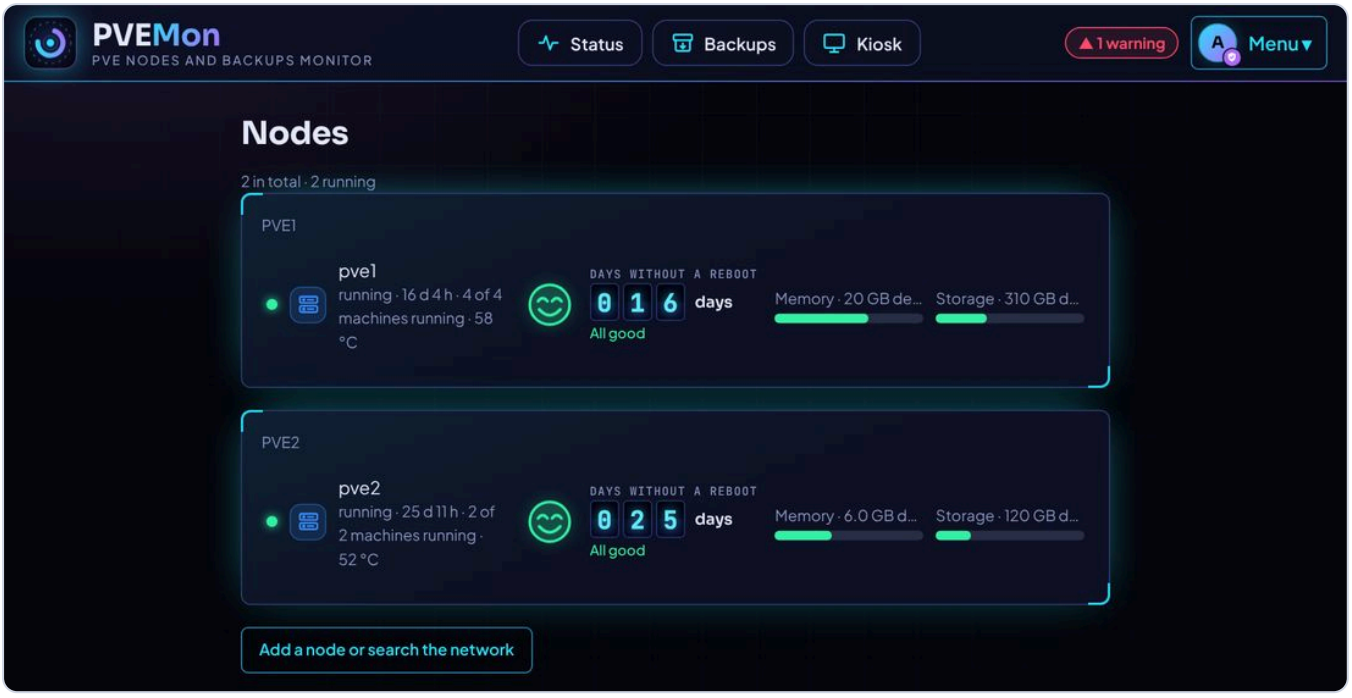
- PVEMon only **reads**; it changes nothing on your servers.
- A user with the **monitor** role only sees and receives notices about what the administrator marks as visible; they cannot enter Settings.
- PVEMon shows you no passwords, keys or internal addresses.
- Important actions are recorded in an administrator audit log.

Administrators' appendix

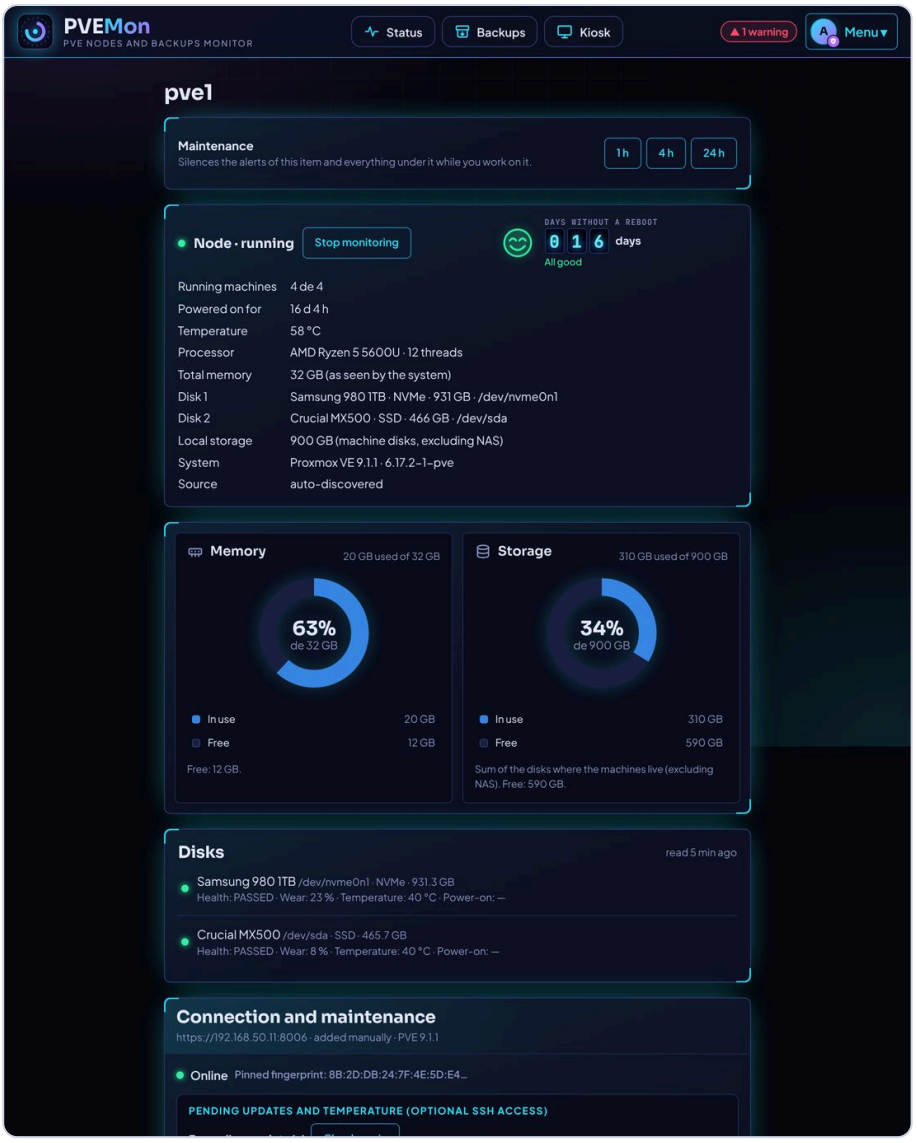
This part is only useful to whoever administers the application; a user with the monitor role does not see these screens.

Nodes, machines and Docker

Nodes is a list with state, uptime, running machines, temperature and memory and storage bars. Clicking a node opens its sheet: machine data, memory and storage rings, **Disks** (health, wear and temperature), **Connection and maintenance** and charts.

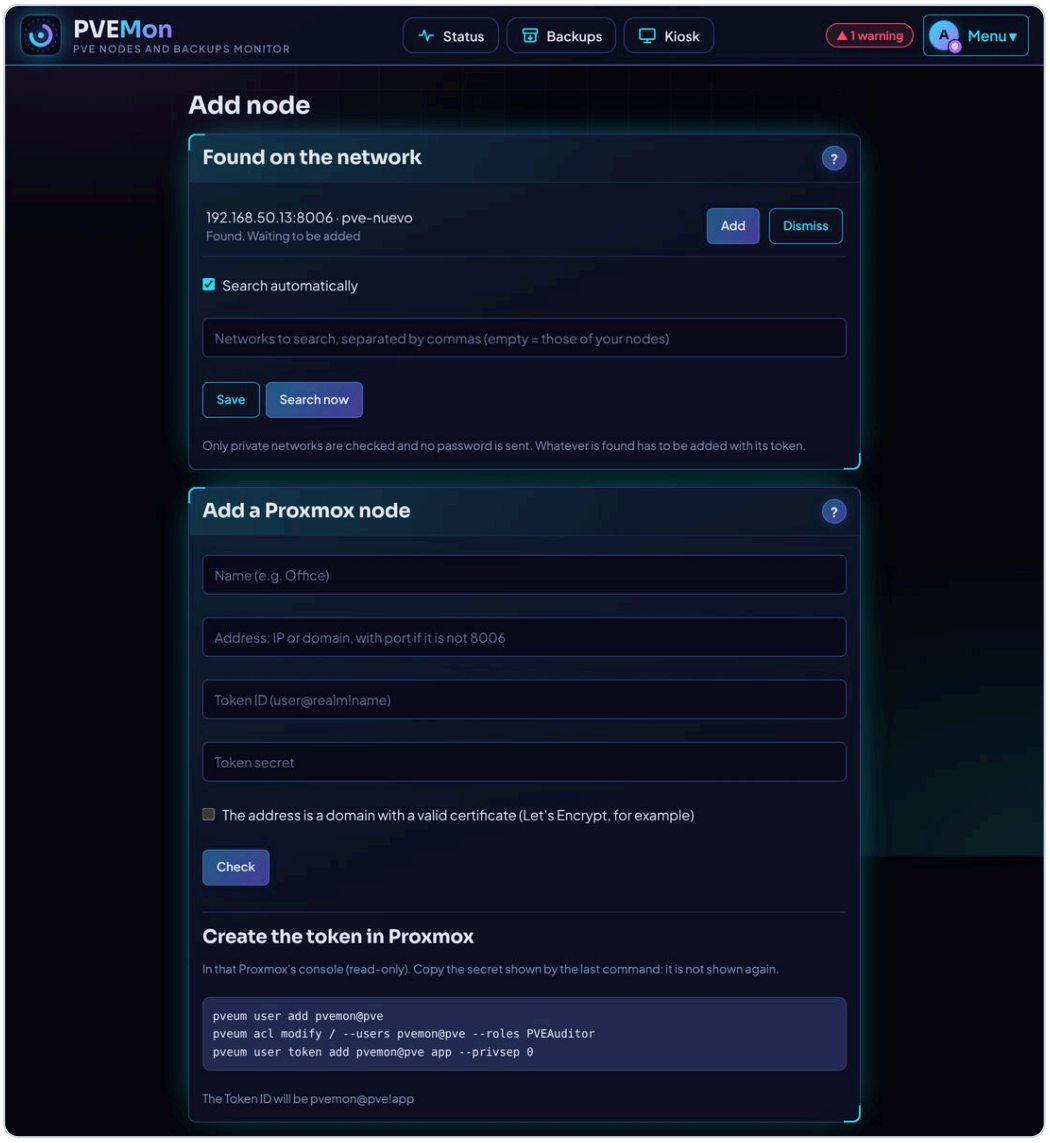


Node list



Node sheet

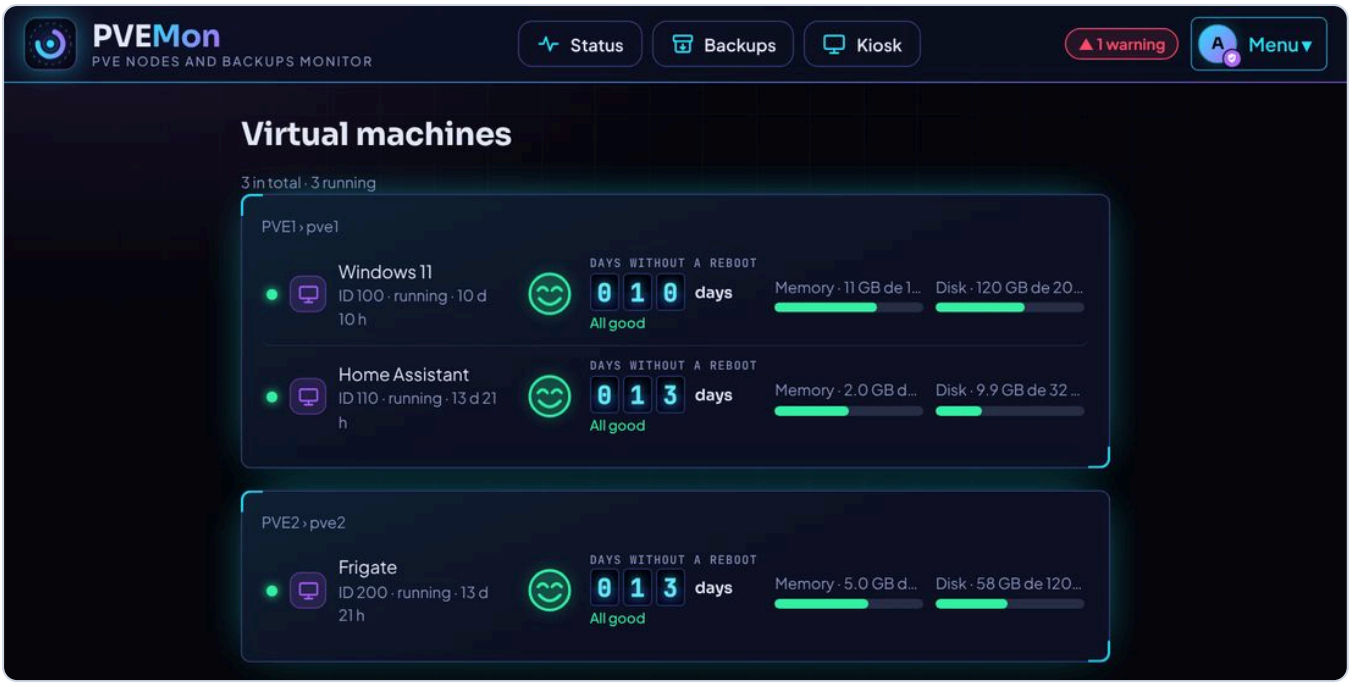
To add a node press **Add node or search the network**: PVEMon looks for Proxmox on your network, or you add it by hand with its address and a **read-only** API token (the screen shows the commands to create it).



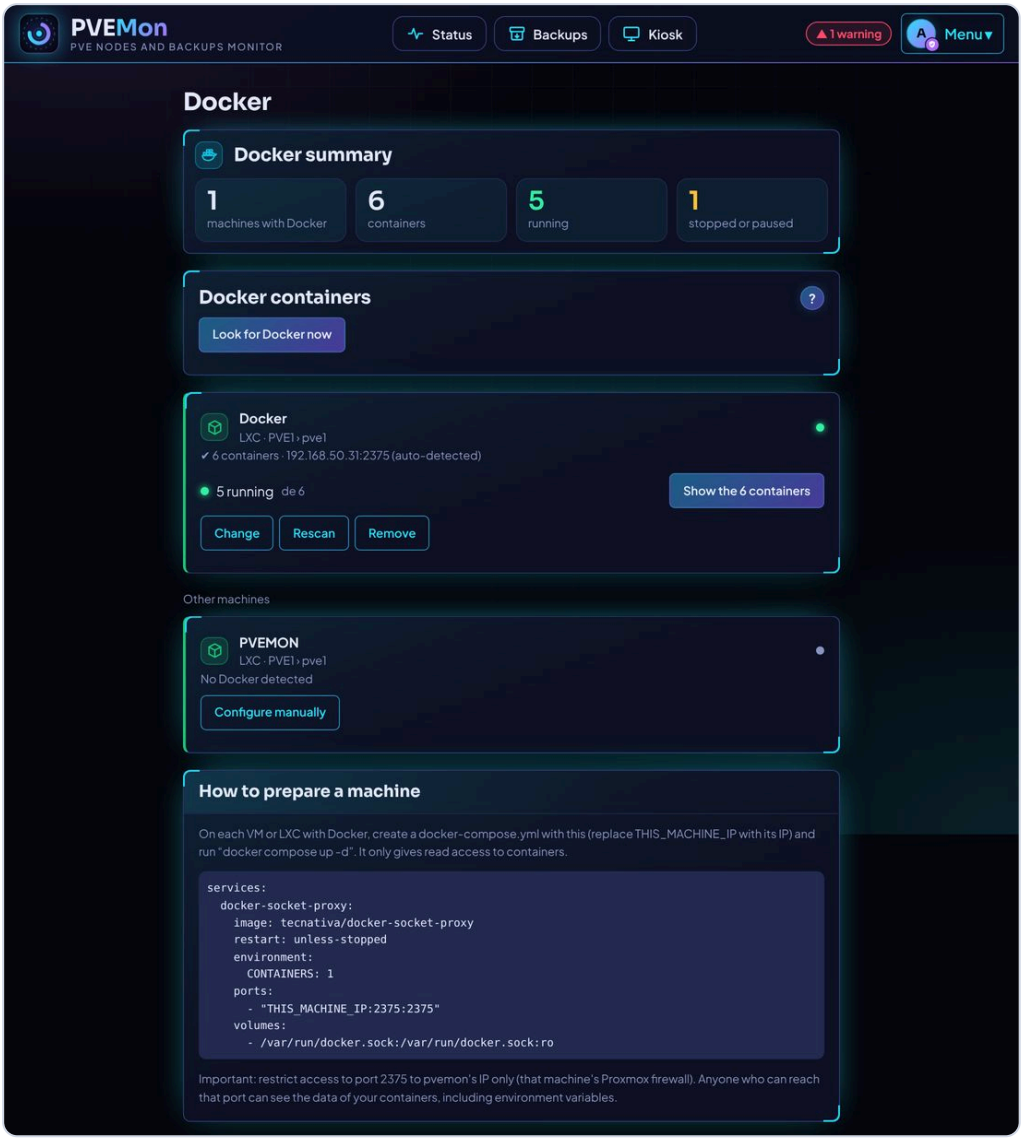
Add node

> PVEMon only reads. It installs no updates and restarts nothing; it only tells you how many are pending. The optional SSH access («Prepare SSH access» block) is limited to three read-only commands: check, temp and version.

VM and **LXC** are identical lists, grouped by node, with memory and disk bars. **Docker** summarises the containers of each Docker host. On a node, VM or LXC sheet, the **1 h, 4 h and 24 h** buttons enable **maintenance mode**, which silences its alerts.



VM list



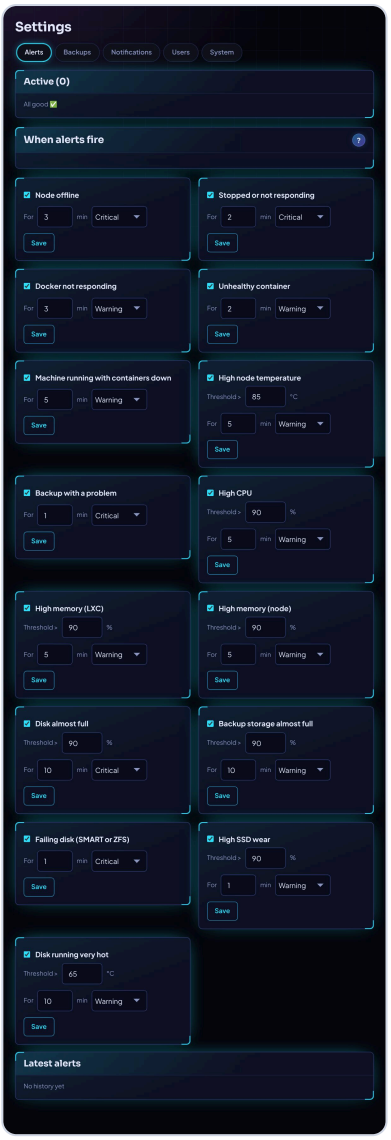
Docker

Settings (administrators)

Settings is divided into tabs. Some have second-level tabs of their own (for example Backups and System).

Tab	What it is for
Alerts	Active and snoozed, when they fire (threshold, duration and severity) and history.
Backups	**PVE backups** (Proxmox backups per server) and **Other backups** (e-mail, files on a network drive).
Notifications	E-mail (SMTP), **weekly summary**, Telegram, webhook and Microsoft Teams.
Users	Accounts, roles, passwords, 2FA and access to each element.
System	Export to BCKMon, external services (heartbeat and API) and PVEMon's backup to a destination.

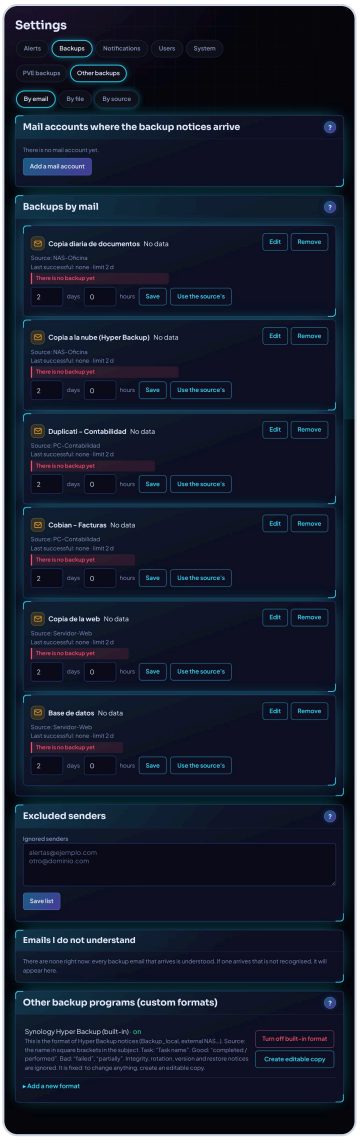
Alerts



Settings › Alerts

Each rule (node offline, high CPU, high memory, failing disk, backup problem…) has its threshold, duration and severity, and a Save button. Each active alert can be **snoozed** (for example 1 hour): it does not warn or count as active and, if it still fails when the time is up, you are warned again.

Backups: backups by e-mail and by file



Settings › Backups › Other backups

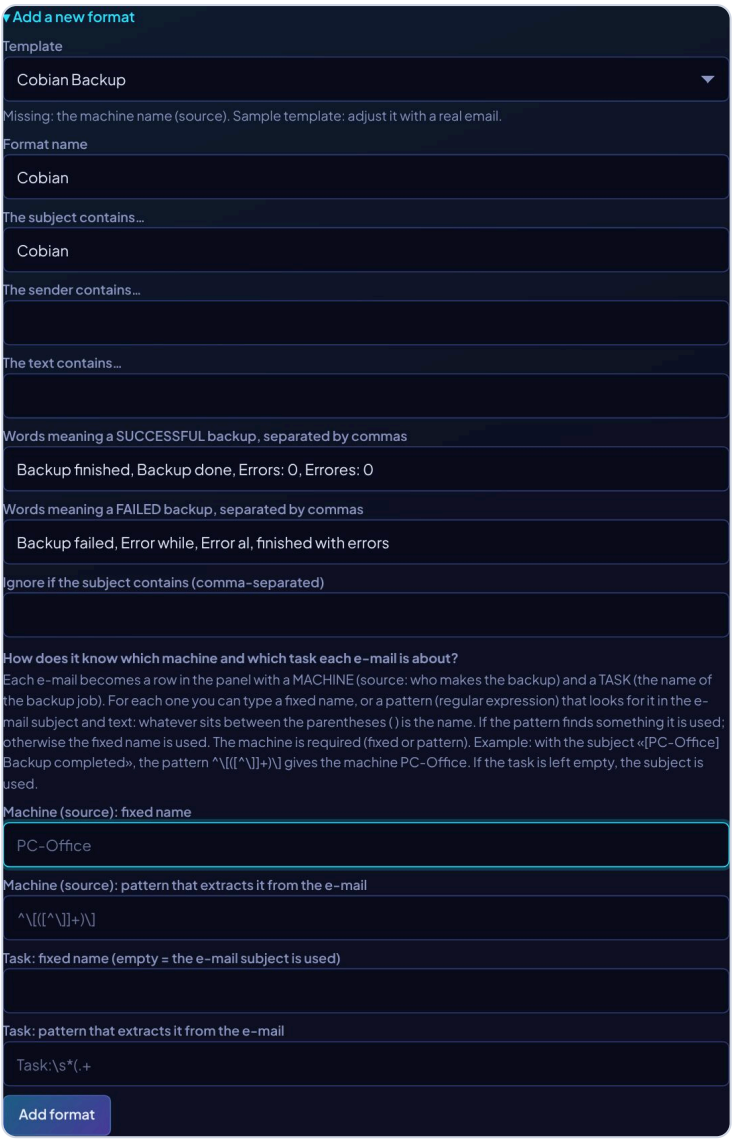
For programs that only report by e-mail (Synology Hyper Backup, Duplicati, Cobian, Proxmox…) you configure an **e-mail account (IMAP)** where the notices arrive; each backup is tied to a **source** (the machine or NAS) and you set how many days may pass without a successful backup. Senders already handled elsewhere can be excluded, and e-mails that are not understood are listed separately.

For what does not report, the **«Check a file or folder on a network drive»** check looks at the NAS directly over NFS: you pick the server, browse the exported folders, set the maximum age and minimum size, and it is checked every 5 minutes. PVEMon speaks NFS version 3 and only reads size and date.

Custom e-mail formats: how templates work

If a backup program is not Synology, you have to tell PVEMon how to recognise its e-mails. In **«Add a new format»** you can start from a **template** (Duplicati, Cobian, Synology, Proxmox) or a blank one, and fill in:

- **When it belongs to this program**: the subject, sender or text contains a word (for example «Cobian»).
- **CORRECT / FAILED words**: separated by commas; if the e-mail contains any failure word the backup counts as failed.
- **Ignore if the subject contains**: for e-mails that are not backups (integrity notices, for example).



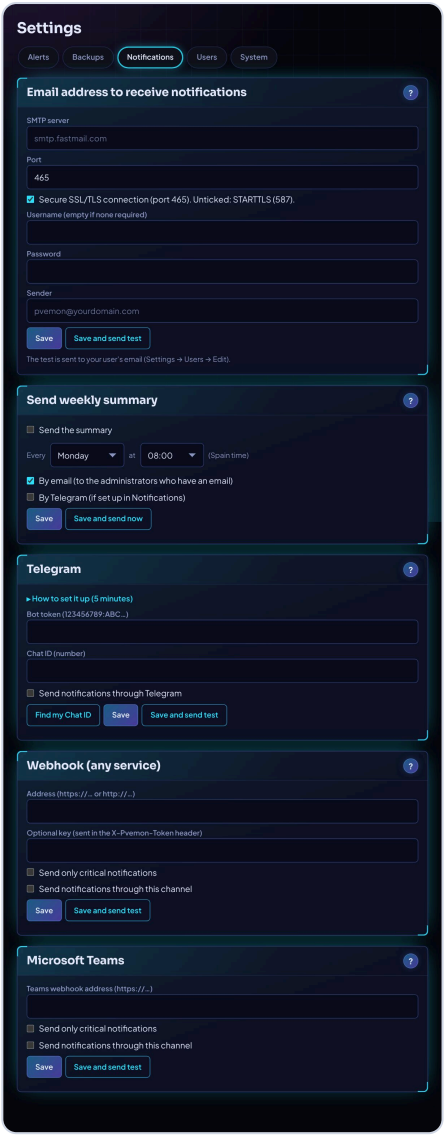
New format form, with the machine and task explanation

Each e-mail becomes a row in the panel with a **machine** (the source: who makes the backup) and a **task** (the name of the backup job). For each there are two fields:

Field	When to use it	Example
Machine: fixed name	All e-mails of this format come from the same machine.	`PC-Office`
Machine: pattern	The machine name is inside the e-mail and changes. A **pattern** (regular expression) looks for it: whatever is **between the parentheses ( )** is the name.	`^\[([^\]]+)\]` extracts «NAS» from «[NAS] Backup completed»
Task: fixed name	One task for all e-mails. Empty = the subject is used.	`Proxmox backup`
Task: pattern	The task name appears in the e-mail.	`Task:\s*(.+)` extracts «Documents» from «Task: Documents»

> If the pattern finds something it is used; otherwise the fixed name is used. The **machine is required** (fixed or pattern). Test the template with a real e-mail from your program before relying on it.

Notifications

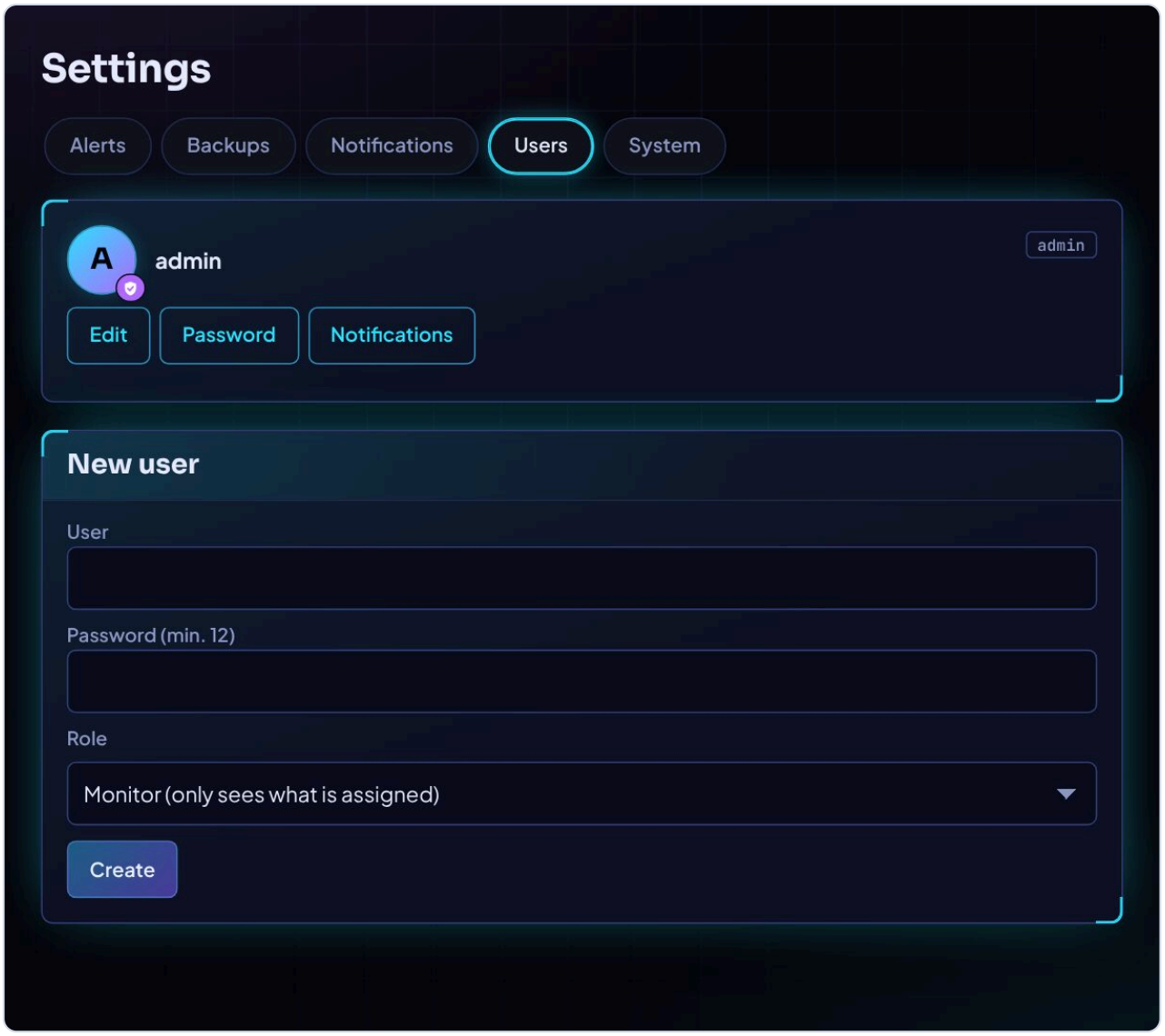


Settings › Notifications

On this tab, from top to bottom: the **e-mail (SMTP)** the notices are sent from; **Send weekly summary**; **Telegram** (bot and chat); a generic **webhook** (a JSON with the alert, with an optional key in the `X-Pvemon-Token` header) and **Microsoft Teams** (a Workflows flow webhook). Each channel has a test button and its addresses and keys are stored encrypted.

Send weekly summary: one message per week with the state of all backups (last successful, size, time to alarm), storage space and open alerts. You choose the day and hour, and whether it goes by e-mail (to administrators with an address) and/or Telegram. «Save and send now» tests it.

Users



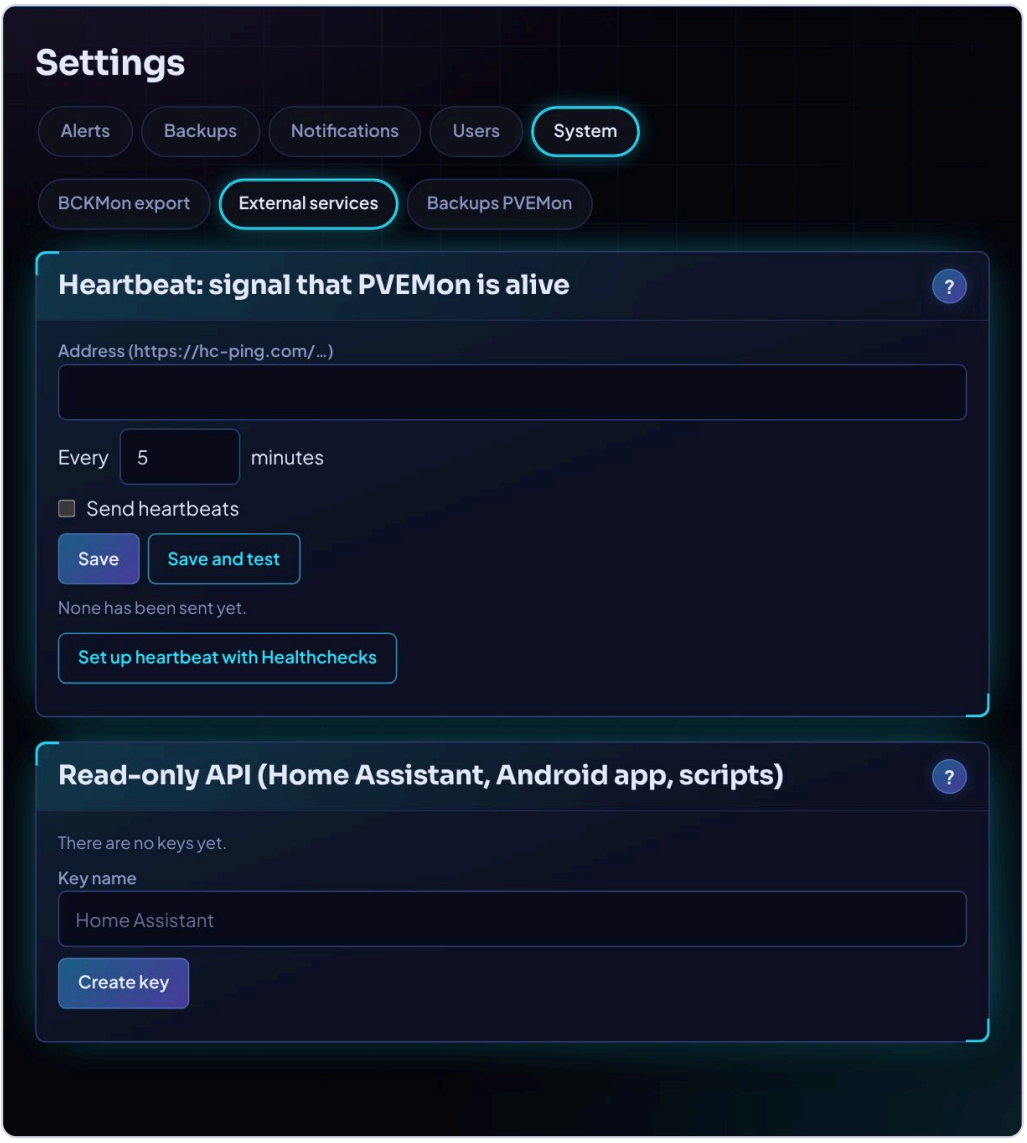
Settings › Users

Here you create accounts and choose their role (**administrator** or **monitor**). A monitor only sees and receives notices about what you mark as visible. The administrator can reset other users' passwords and 2FA.

System

Three second-level tabs:

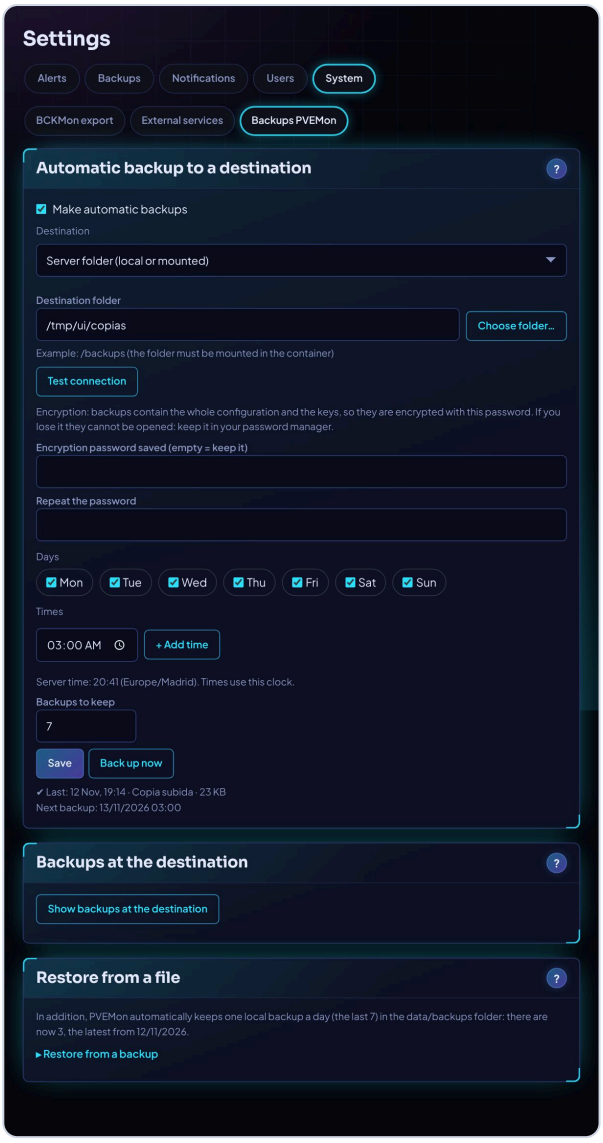
- **BCKMon export**: creates an encrypted file with only the backup configuration (sources, backups, e-mail, excluded senders and file checks; no users, kiosks or alerts) to import into BCKMon; and brings here what was exported from BCKMon, adding only what is missing.
- **External services**: the **heartbeat** (visits an address every N minutes — Healthchecks.io, Uptime Kuma, Home Assistant — so an outside service warns you if the application stops working) and the **read-only API** (keys for Home Assistant or scripts; shown once and unable to change anything).
- **PVEMon backups**: the automatic backup of the application itself (see below).



Settings › System › External services

Automatic PVEMon backup to a destination

The **PVEMon backups** tab makes an encrypted copy of the configuration (users, settings, keys) and leaves it at the destination you choose, on the days and times you want, keeping the last N. The copy is encrypted with a **passphrase you choose** (minimum 8 characters): if you lose it the copy cannot be opened.



Settings › System › PVEMon backups

Destination	What you need
Server folder	A folder (local or mounted) visible from the container, for example `/backups`.
NFS server	Server and exported folder (NFSv3) with write permission for the application's IP.
SMB/CIFS	Server, share, user and password.
SFTP	Server, user and password, and the folder.
Dropbox / Google Drive / OneDrive	It connects with an authorisation from the cloud itself (**Connect** button); your username and password are not stored.

The **Test connection** and **Choose folder...** buttons check the destination before saving; **Copy now** makes a copy immediately and **Copies at the destination** lists those present and lets you **Restore**. Restoring replaces the current configuration, logs everybody out and restarts the application; it only works with the same secret key (`SECRET_KEY`) it was made with. The copy appears in the Backups panel, and an alert fires if a scheduled one is not made.

> Keep your `SECRET_KEY` and the backup passphrase in your password manager: without them a restore is not possible.

Security and resources

- Passwords with scrypt, optional 2FA and account lock after 10 failed attempts in 15 minutes (plus a per-IP limit).
- HttpOnly + SameSite=Strict session cookie (and Secure over HTTPS); it expires after 14 days.
- Request origin checks, security headers and generic errors.
- Keys, tokens and passwords stored encrypted with `SECRET_KEY`.
- Audit log of important actions.

Recommendations

- Publish PVEMon only over HTTPS (for example with Nginx Proxy Manager) and close port 3000 to the rest of the network.
- Enable 2FA for all users.
- Keep `SECRET_KEY` and the `data` folder in your backups.

Server resources

Resource	Need
Processor	1 core (almost no load)
Memory	~180 MB idle; reserve 512 MB
Disk	Image ~400 MB plus the data
Network	In: port 3000. Out: e-mail, Telegram, push, webhooks, NFS/SMB/SFTP or backup clouds, nodes (8006) and Docker (2375)

Recent changes

The full history is under the **ChangeLog** button in the menu.

Version	Change
2.3.x	Settings cards with a title band; «Backups by mail» in its own card; backup sheet with a face and a counter towards the alarm; Theme first in the menu. Header with centred navigation; reordered menu (Log out after the options, Theme last on one line, manual link); weekly summary inside Notifications («Send weekly summary»); alerts in a single card; better-explained e-mail formats; application backup first in Backups; updated PDF guides and the first one for BCKMon.
2.2.x	Automatic application backup to folder, NFS, SMB, SFTP, Dropbox, Google Drive and OneDrive; Settings reorganised with second-level tabs; the own backup shows in the Backups panel.
2.0 – 2.1	BCKMon edition (backups only), five colour palettes and a name per edition.
1.5.x	Spanish and English; Docker image; PDF guide.
1.x	First stable version, read-only API, multiple kiosks, disks, webhook and Teams, 2FA.